AMS 526: Numerical Linear Algebra for Computational and Data Sciences

Lecture 21: Arnoldi and Lanczos Iterations

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Outline

1. **Krylov Subspace and Arnoldi Iterations (NLA§32-33)**
2. Lanczos Iterations (NLA§36)

Krylov Subspace Methods

- Given an $n \times n$ matrix A and a nonzero vector b, the k-th Krylov subspace is defined as

$$\mathcal{K}_k(A, b) = \text{span}\{b, Ab, A^2 b, \dots, A^{k-1} b\}$$

	linear systems	eigenvalue problems
Hermitian	CG	Lanczos
Nonhermitian	GMRES, BiCG, etc.	Arnoldi

- CG, GMRES, etc., are Krylov subspace methods for solving sparse linear systems (later).
- Lanczos and Arnoldi iterations are Krylov subspace methods for reduction to tridiagonal or Hessenberg form, respectively.

Review: Reduction to Hessenberg Form

- General A: First convert to *upper-Hessenberg* form, then to upper triangular

$$\underset{A \neq A^*}{\begin{bmatrix} \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \end{bmatrix}} \xrightarrow{\text{Phase 1}} \underset{\text{upper-Hessenberg}}{\begin{bmatrix} \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ & \times & \times & \times & \times \\ & & \times & \times & \times \\ & & & \times & \times \end{bmatrix}} \xrightarrow{\text{Phase 2}} \underset{\text{triangular}}{\begin{bmatrix} \times & \times & \times & \times & \times \\ & \times & \times & \times & \times \\ & & \times & \times & \times \\ & & & \times & \times \\ & & & & \times \end{bmatrix}}$$

- Hermitian A: First convert to *tridiagonal* form, then to diagonal

$$\underset{A = A^*}{\begin{bmatrix} \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \end{bmatrix}} \xrightarrow{\text{Phase 1}} \underset{\text{tridiagonal}}{\begin{bmatrix} \times & \times & & & \\ \times & \times & \times & & \\ & \times & \times & \times & \\ & & \times & \times & \times \\ & & & \times & \times \end{bmatrix}} \xrightarrow{\text{Phase 2}} \underset{\text{diagonal}}{\begin{bmatrix} \times & & & & \\ & \times & & & \\ & & \times & & \\ & & & \times & \\ & & & & \times \end{bmatrix}}$$

- In general, Phase 1 is direct and requires $\mathcal{O}\left(n^3\right)$ flops.

Arnoldi Iteration

- The Arnoldi iteration reduces a general, nonsymmetric matrix A to Hessenberg form by similarity transformation $A = QHQ^*$.
- It is analogous to a Gram-Schmidt-style iteration instead of Householder reflections.
- Let $Q_k = [q_1 \mid q_2 \mid \cdots \mid q_k]$ be an $n \times k$ matrix with the first k columns of Q, and $\tilde{H}_k$ be the $(k+1) \times k$ upper-left section of H, i.e., $\tilde{H}_k = H_{1:k+1,1:k}$.
- Consider the first k columns of $AQ = QH$, or $AQ_k = QH_{:,1:k} = Q_{k+1}\tilde{H}_k$:

$$\underbrace{\begin{bmatrix} & & \\ & A & \\ & & \end{bmatrix}}_{A} \underbrace{\left[\begin{array}{c|c|c} q_1 & \cdots & q_k \end{array}\right]}_{Q_k} = \underbrace{\left[\begin{array}{c|c|c} q_1 & \cdots & q_{k+1} \end{array}\right]}_{Q_{k+1}} \underbrace{\begin{bmatrix} h_{11} & \cdots & h_{1k} \\ h_{21} & & \\ & \ddots & \vdots \\ & & h_{k+1,k} \end{bmatrix}}_{\tilde{H}_k}$$

- **Question**: How do we choose q_1?

Arnoldi Algorithm

- Start with a random q_1, then determine q_2 and $\tilde{H}_1$, and so on.
- The kth column of $AQ_k = Q_{k+1}\tilde{H}_k$ can be written as

$$Aq_k = h_{1k}q_1 + \cdots + h_{kk}q_k + h_{k+1,k}q_{k+1}$$

where $h_{jk} = q_j^* A q_k$.

Algorithm: Arnoldi Iteration

given random nonzero b, let $q_1 = b/\|b\|$
for $k = 1, 2, 3, \ldots$
    $v = Aq_k$
    for $j = 1$ **to** k
        $h_{jk} = q_j^* v$
        $v = v - h_{jk}q_j$
    $h_{k+1,k} = \|v\|$; $q_{k+1} = v/h_{k+1,k}$

- **Question**: What if q_1 happens to be an eigenvector?

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- **Question**: What if q_1 happens to be an eigenvector?
- **Answer**: $h_{2,1} = 0$; continue with an arbitrary unit vector q_2 orthogonal to q_1.

QR Factorization of Krylov Matrix

- Assuming exact arithmetic and $h_{k+1,k} \neq 0$, the vectors $q_1, \dots, q_k$ from Arnoldi form an orthonormal basis of the successive Krylov subspaces

$$\mathcal{K}_k = \operatorname{span}\left\{b, Ab, \dots, A^{k-1}b\right\} = \operatorname{span}\left\{q_1, q_2, \dots, q_k\right\} \subseteq \mathbb{C}^n$$

- Q_k comes from the reduced QR factorization $K_k = Q_k R_k$ of the Krylov matrix

$$K_k = \left[\begin{array}{c|c|c|c} b & Ab & \cdots & A^{k-1}b \end{array}\right]$$

- However, K_k and R_k are not formed explicitly; forming them explicitly would be highly ill-conditioned and unstable.

Projection onto Krylov Subspaces

- In exact arithmetic, the Arnoldi process computes orthogonal projections of A onto successive Krylov subspaces
$$H_k = Q_k^* A Q_k$$
because $AQ_k = Q_{k+1}\tilde{H}_k$, $\tilde{H}_k = Q_{k+1}^* A Q_k$, and $H_k = \tilde{H}_{1:k,1:k}$.
- H_k can be interpreted as the orthogonal projection of A onto $\mathcal{K}_k$ in the basis $\{q_1, q_2, \dots, q_k\}$, restricting the mapping $A : \mathbb{C}^n \to \mathbb{C}^n$ to $H_k : \mathcal{K}_k \to \mathcal{K}_k$. This kind of projection is known as a *Rayleigh-Ritz procedure*.
- Arnoldi iteration is useful as:
  1. a basis for iterative algorithms (such as GMRES, to be discussed later).
  2. a technique for estimating eigenvalues of nonhermitian matrices.
- **Caution**: Eigenvalues of nonnormal matrices may have little to do with the physical system, since eigenvalues of such matrices can be highly ill-conditioned.

Estimating Eigenvalues by Arnoldi Iteration

- Diagonal entries of H_k are Rayleigh quotients of A w.r.t. vectors q_i.
- H_k is a "generalized Rayleigh quotient" w.r.t. Q_k, whose eigenvalues $\{\theta_j\}$ are called *Arnoldi estimates* or *Ritz values* w.r.t. $\mathcal{K}_k$ of A.
- *Ritz vectors* corresponding to θ_j are $Q_k y_j$, where $H_k y_j = \theta_j y_j$.
- To use Arnoldi iteration to estimate eigenvalues, compute the eigenvalues of H_k at the kth step.
- In exact arithmetic, when $k = n$, the Ritz values are the exact eigenvalues.
- In general, $k \ll n$, so we can estimate only a few eigenvalues.
- Typically, the iteration finds extreme eigenvalues first, which are often of main interest in applications:
  - Stability analysis typically requires estimating the spectral radius.
  - Principal Component Analysis (PCA) requires estimating the largest eigenvalues and corresponding eigenvectors of $A^\top A$.

Invariance Properties of Arnoldi Iteration

Theorem

Let the Arnoldi iteration be applied to matrix $A \in \mathbb{C}^{n\times n}$ as described above.

Translation invariance. If A is changed to $A + \sigma I$ for some $\sigma \in \mathbb{C}$, and b is unchanged, then Ritz values $\{\theta_j\}$ change to $\{\theta_j + \sigma\}$.

Scale invariance. If A is changed to σA for some $\sigma \in \mathbb{C}$, and b is unchanged, then $\{\theta_j\}$ change to $\{\sigma\theta_j\}$.

Unitary similarity invariance. If A is changed to UAU^* for some unitary matrix U, and b is changed to Ub, then $\{\theta_j\}$ do not change.

In all three cases, the Ritz vectors, namely $Q_k y_j$ corresponding to eigenvectors y_j of H_k, do not change under the indicated transformation.

Convergence of Arnoldi Iteration

- In exact arithmetic, if A has n distinct eigenvalues, the Arnoldi iteration finds them all in at most n steps.
- Under certain circumstances, convergence of some Arnoldi estimates is geometric (i.e., linear), and it accelerates in later iterations.
- Convergence depends heavily on the spectrum and the nonnormal properties of A.

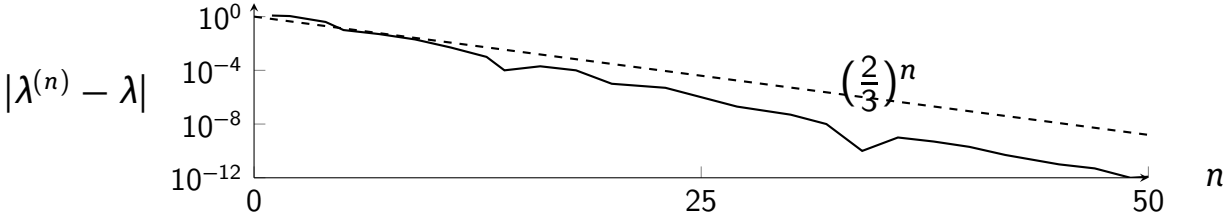

Figure: Example convergence of extreme Arnoldi eigenvalue estimation.

Outline

1. Krylov Subspace and Arnoldi Iterations (NLA§32-33)
2. **Lanczos Iterations (NLA§36)**

Lanczos Iteration for Symmetric Matrices

- For symmetric (or Hermitian) A, the matrices $\tilde{H}_k$ and H_k in the Arnoldi iteration become tridiagonal.
- We denote them by $\tilde{T}_k$ and T_k, respectively. Let $\alpha_k = h_{kk}$ and $\beta_k = h_{k+1,k} = h_{k,k+1}$.
- $AQ_k = Q_{k+1}\tilde{H}_k$ can then be written as a three-term recurrence

$$Aq_k = \beta_{k-1}q_{k-1} + \alpha_k q_k + \beta_k q_{k+1}$$

where α_i are diagonal entries and β_i are sub-diagonal entries of $\tilde{T}_k$

$$T_k = \begin{bmatrix} \alpha_1 & \beta_1 & & & \\ \beta_1 & \alpha_2 & \beta_2 & & \\ & \beta_2 & \alpha_3 & \ddots & \\ & & \ddots & \ddots & \beta_{k-1} \\ & & & \beta_{k-1} & \alpha_k \end{bmatrix}$$

- The Arnoldi iteration specialized for symmetric matrices is known as the Lanczos iteration.

Algorithm of Lanczos Iteration

Algorithm: Lanczos Iteration

$\beta_0 = 0,\ q_0 = 0$
given random b, let $q_1 = b/\|b\|$
for $k = 1, 2, 3, \ldots$
$\quad v = Aq_k$
$\quad \alpha_k = q_k^* v$
$\quad v = v - \beta_{k-1} q_{k-1} - \alpha_k q_k$
$\quad \beta_k = \|v\|;\ q_{k+1} = v/\beta_k$

- Each step requires only one matrix-vector multiplication, two inner products, and a few vector updates, making it highly efficient for sparse matrices.
- Like Arnoldi, the Lanczos iteration is highly useful as:
  1. a basis for other iterative algorithms (e.g., Conjugate Gradient).
  2. a technique for estimating eigenvalues of Hermitian matrices.

Estimating Eigenvalues by Lanczos Iteration

- For symmetric matrices with evenly spaced eigenvalues, Ritz values tend to first converge to the extreme eigenvalues.

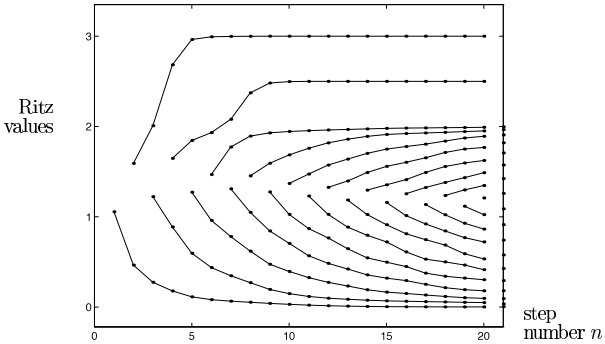

Figure: Ritz values for first 20 steps for Lanczos iteration applied to example 203×203 matrix. Convergence of extreme eigenvalues is geometric.

Effect of Rounding Errors

- Rounding errors have complex effects on the Lanczos iteration and all iterations based on three-term recurrences.
- In finite precision, rounding errors cause a loss of orthogonality among $q_1, q_2, \dots, q_k$:
  - In Arnoldi, vectors $q_1, \dots, q_k$ are explicitly orthogonalized via modified Gram-Schmidt, which suffers some, but not total, loss of orthogonality.
  - In Lanczos, orthogonality of q_j with q_{j-1} and q_{j-2} is explicitly "enforced", but orthogonality with $q_{j-3}, \dots, q_1$ is only "automatic" in exact arithmetic.
  - In practice, these mathematical identities are not accurately preserved, leading to a rapid loss of global orthogonality.
- Periodic full or selective re-orthogonalization of Q_k is often used to alleviate the effect of rounding errors.

Rounding Errors and Ghost Eigenvalues

- Without re-orthogonalization, the Lanczos iteration suffers from loss of orthogonality, which can in turn lead to spurious "ghost" eigenvalues.

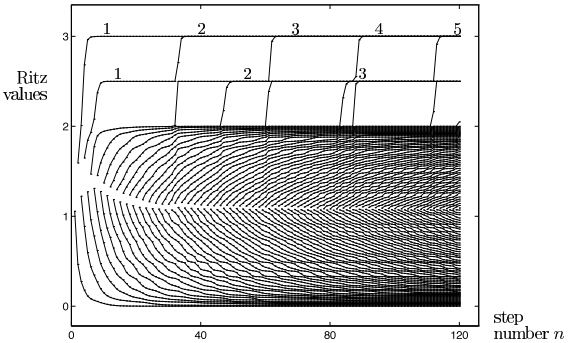

Figure: Continuation to 120 steps of Lanczos iteration. Numbers indicate multiplicities of Ritz values. 4 "ghost" copies of 3.0 and 2 "ghost" copies of 2.5 appear.

Explanation of Ghost Eigenvalues

- Intuitive explanation of ghost eigenvalues:
  - Convergence of a Ritz value annihilates the corresponding eigenvector components in the residual vector.
  - However, random noise from rounding errors re-introduces and excites those components again, causing the algorithm to "rediscover" the same eigenvalue.
- We cannot trust the multiplicities of Ritz values as the true algebraic multiplicities of the eigenvalues.
- Nevertheless, the Lanczos iteration remains very useful in practice:
  - E.g., in PCA for dimensionality reduction, one needs to find the leading singular values and corresponding singular vectors of A.
  - One standard approach is to apply the Lanczos iteration implicitly to $A^\top A$ or $AA^\top$ without forming the product explicitly, and then use the Ritz vectors to approximate the singular vectors.